Reg. No. | | | | | | | | | | | [14ME72]

SEVENTH SEMESTER B.Tech EXAMINATION, DECEMBER–2018

OPERATIONS RESEARCH

Time:3 Hours **Maximum Marks:70**

Instructions: i. Answer any **FIVE** full Questions choosing atleast two from each part.

ii. Missing data may be suitably assumed.

PART - A

1. a. Define operations research (O R) and explain briefly the various phases of O R? **06**

b. A Production planner in a soft drink plant with two bottling machines A and B has given the following information. A is designed for 8 ounce bottles and B is designed for 16 ounce bottles. However each can be used on both types of bottles with some loss of efficiency. The following information is available: **08**

Machine	8–Ounce Bottles	16–Ounce Bottles
A	100/Minute	40/Minute
B	60/Minute	75/Minute

The machines can be run for 8 hours per day, 5 days a week. Profit on 8-ounce bottle is 15 paise and on 16-ounce bottle is 25 paise. Weekly production of the drink cannot exceed 3,00,000 ounces and the market can absorb only 25,000 eight-ounce bottles and 7000 Sixteen-ounce bottles per week. The planner wishes to maximize his profit subject to the above constraints. Formulate the problem as LPP.

2. a. Define the following terms with suitable example with respect to SIMPLEX METHOD **04**

i. SURPLUS VARIABLE

ii. ARTIFICIAL VARIABLE

b. Solve the following L P P by using SIMPLEX Method **10**

Maximize $Z = 2x1 + 2x2$

subjected to $5x1 + 3x2 \leq 8$

$2x1 + 4x2 \leq 8$

and $x1, x2 \geq 0$

3. a. A manufacturing company has 3 factories A,B and C which supplies products to warehouses D,E,F and G. Monthly capacities for factories A, B and C are 250, 300 and 400 units for regular production. If overtime is utilized factories A and B can produce 50 and 75 additional units. The unit incremental costs for overtime are Rs. 4 and Rs. 5 respectively. The current ware house requirements are 200, 225, 275 and 300 units for D, E, F and G respectively. The unit transportation costs are given in the table in Rs. **10**

		TO			
FROM		D	E	F	G
	A	11	13	17	14
	B	16	18	14	10
	C	21	24	13	10

Determine the optimum distribution schedule which satisfies requirements of all warehouses.

b. Explain briefly how to overcome degeneracy in transportation problem. **04**

4. a. A College athletic conference has six basket ball officials and it must assign to 3 conference games. Two officials must be assigned to each game. The conference office desires to assign the officials such that the total distance travelled by all the six officials will be minimized. The distances each official would have to travel to each game are given below. **10**

		GAME		
OFFICIAL		A	B	C
	1	20	45	10
	2	40	90	70
	3	60	70	30
	4	30	60	40
	5	70	15	50
	6	80	25	35

The conference office has decided not to assign official 4 to game A because of previous conflicts with one of the coaches. Determine optimum assignment.

b. List the differences between transportation problem and assignment Problem. **04**

PART - B

5. a. Determine the minimum elapsed time for the two jobs for which details of processing times and sequence of operations are as given below: **08**

JOB 1: A – 6 to C – 7 to B – 5 to D – 4 to E – 5

JOB 2: A – 4 to B – 3 to C – 5 to E – 2 to D – 4

b. Find the sequence that minimizes the total processing time required in performing the jobs on 3 machines in the order C B A **06**

		MACHINE		
JOB		A	B	C
	1	8	3	8
	2	7	4	3
	3	6	5	7
	4	9	2	2
	5	10	1	5
	6	9	6	1

6. a. Solve the following Game by principle of Dominance **06**

		PLAYER B			
		B1	B2	B3	B4
PLAYER A	A1	3	2	4	0
	A2	3	4	2	4
	A3	4	2	4	0
	A4	0	4	0	8

b. Solve the following game by using Graphical Method **08**

		PLAYER B			
		B1	B2	B3	B4
PLAYER A	A1	2	1	0	-2
	A2	1	0	3	2

7. a. List the basic differences in PERT and CPM with respect to networks. **04**

b. For the following activities construct the network, number the events, find the critical path and its duration; **10**

Sl No	JOB	PREDECESSOR	DURATION
1	A	-	3
2	B	-	6
3	C	-	5
4	D	A	5
5	E	B	4
6	F	B	6
7	G	B	3
8	H	C, G	4
9	I	D, E	8
10	J	D, E	4
11	K	J, F, G, C	6
12	L	J, F, G, C	5
13	M	I, K	3
14	N	L, M,H	7

8. a. With suitable examples briefly explain the following terms with respect to Queuing Theory **06**

i. BALKING

ii. JOCKEYING

iii. RENEGING

b. In a railway yard, goods trains arrive at a rate of 30 trains per day. Assume that the inter arrival time follows an exponential distribution and the service time distribution also follows exponential distribution with an average of 36 minutes. Calculate the following: **08**

i. The average number of trains in the system

ii. The probability that the total queue size exceeds 10

iii. Average waiting time